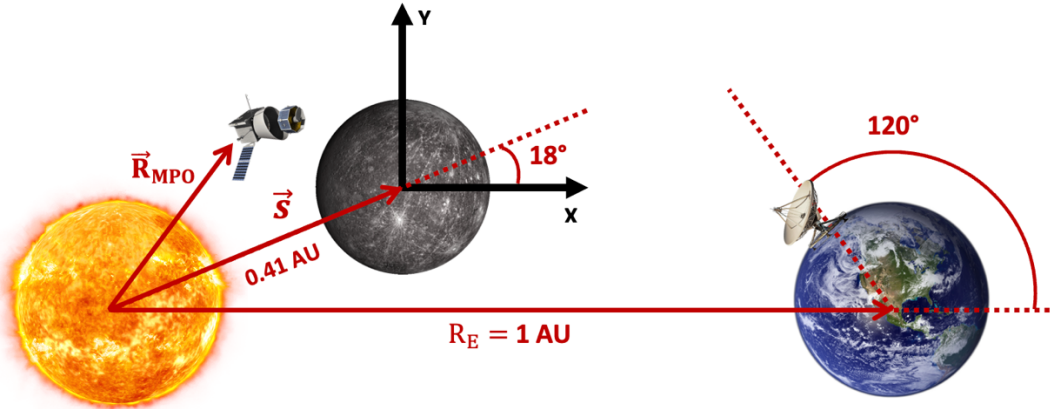


Space Missions and Systems 2026: Homework 1

The Mercury Planetary Orbiter (MPO) will be captured by Mercury's gravity in November 2026. It will be tracked by a ground station providing range and range-rate measurements. For this homework, let us assume that the spacecraft orbit is in the plane containing the Earth, Mercury and the Sun (2D problem). The Earth and Mercury are considered fixed in space and non-rotating. Consider a reference frame centered in the center of mass of the planet.

The figure below illustrates the details of the geometry at the reference epoch ($t_0 = 0$):



The spacecraft is subject to the monopole gravity of Mercury and to the solar radiation pressure.

Assume that the acceleration due to solar pressure is described by:

$$\vec{a}_{\text{SRP}}(t) = \frac{\Phi}{c} \cdot \left(\frac{1 \text{ AU}}{R_{\text{MPO}}(t)} \right)^2 \cdot \frac{A}{m} \cdot (1 + C_s) \hat{s}$$

where Φ is the solar flux at 1AU, c is the speed of light, $R_{\text{MPO}}(t)$ is the instantaneous distance between the Sun and the spacecraft, A is the effective cross-sectional area of the spacecraft, m is the mass of the spacecraft, and C_s its specular reflection coefficient. Assume that the direction from the Sun to Mercury, $\hat{s}$, is fixed. Do not consider solar eclipses (solar radiation pressure is supposed to act continuously).

The a priori values of the state vector with respect to Mercury at the reference epoch ($t_0 = 0$) are:

State parameter	A priori value	A priori uncertainty
x [km]	-2840.98	0.1
y [km]	761.22	0.1
v_x [km/s]	-0.853853	1e-5
v_y [km/s]	-2.802925	1e-5
GM_{merc} [km ³ /s ²]	22032.55	0.5
C_s	0.083	0.01

The a priori covariance matrix is assumed to be diagonal. The ground station equipment provides range and range-rate measurements with a putative accuracy of 1 m and 0.1 mm/s.

The observables are provided in the “observables.txt”: the first column contains the observation time (in seconds past t_0), the second column is the corresponding observed range value (in km), and the third column is the corresponding observed range-rate value (in km/s).

- (1) Use the range and range-rate observables provided by the ground station to obtain an estimate of the six state vector's components and the associated uncertainties at the reference epoch. Report the new initial state estimate.
- (2) Which of the two types of observables is the most valuable for the determination of the state vector components? Justify your answer.
- (3) Why do observables present a gap between $t_0 + 7260\text{s}$ and $t_0 + 9600\text{s}$? Justify your answer.
- (4) Plot the evolution in time of the altitude and its uncertainty during the observation period¹.
- (5) Extra: starting from the estimated state at point 1, integrate the spacecraft trajectory from t_0 to the time of the last observable considering solar eclipses. Plot the evolution in time of the difference in altitude between the new trajectory and the reference trajectory obtained at point one (without eclipses). In the same figure plot also the altitude uncertainty in time obtained at point 4. Comment your results.

Read carefully the following information:

- Astronomical Unit (AU): 149597870.70 km
- Solar flux at 1 AU (Φ): 1371 W/m²
- Spacecraft total effective area (A): 5.2 m²
- MPO mass: 1107.25 kg
- Distance between Mercury and the Sun: 0.41 AU
- Distance between the Earth and the Sun: 1.0 AU
- Earth radius: 6378.14 km
- Mercury radius: 2440.0 km
- Speed of light (c): 299792.458 km/s

Upload on Google Classroom a **zip folder** containing a working computer code and instructions for its use. The code must run with a one-line command. It must print tables and produce the relevant plots (MATLAB is recommended). You must provide a **comprehensive and concise note in English** (a pdf file **not included the zip folder**) containing **all** mathematical procedures, results, comments, tables and figures by **Sunday 12 April 23:59**.

Only questions pertaining to the text will be answered. Please ask questions publicly on Classroom (in English).

Tips on Matlab:

- Rather than ode45, use ode113, which excels in orbital dynamics problems (just replace ode45 with ode113 in your setup);
- For integration purposes, use at least RelTol=1.0E-13, AbsTol=1.0E-13;
- ALWAYS put labels and units on the plot axes!

¹For the uncertainty of derived quantities, see https://en.wikipedia.org/wiki/Propagation_of_uncertainty (section "non-linear combinations")